Supplementary File 6. Evidence classification codebook

Review title

Reframing the malaria-sickle cell disease relationship in sub-Saharan Africa as a bounded syndemic: A scoping review and evidence gap map

1. Purpose

This codebook defines the evidence classification system used to code included sources in the scoping review. It provides operational definitions, inclusion rules, exclusion rules, examples, and reviewer notes for each evidence category.

The codebook is designed to ensure consistent classification of evidence across seven syndemic-relevant domains:

1. Co-occurrence
2. Spatial co-risk
3. Biological interaction
4. Clinical convergence
5. Service-mediated visibility
6. Structural vulnerability
7. Programme response

It also provides coding guidance for explicit syndemic theory, implicit syndemic logic, and level of evidence integration.

2. General coding principles

Each included source should be coded into one or more evidence categories. A study does not need to address all categories to be included. It must contribute meaningfully to at least one category.

Reviewers should apply the following principles:

1. Code what the study actually reports, not what it could theoretically imply.
2. Use multiple codes when a source contributes evidence to more than one category.
3. Do not code co-occurrence as syndemic interaction unless the study provides evidence of interaction, convergence, structural amplification, or programme linkage.
4. Distinguish biological interaction from spatial overlap.
5. Distinguish clinical convergence from service-mediated visibility.
6. Distinguish structural vulnerability from programme response.
7. Record reviewer justification for all syndemic coding decisions.
8. Use "unclear" only when the full text does not provide enough information for confident coding.

3. Core evidence categories

Table 1. Summary of coding categories

Code	Category	Core question	Minimum evidence required
C1	Co-occurrence	Are malaria and SCD, HbS, or related	Evidence of shared population,

		red cell traits present in the same population, region, or setting?	geography, or clinical context.
C2	Spatial co-risk	Are malaria exposure and HbS, SCD, SCA births, or service need geographically patterned or overlapping?	Mapping, spatial distribution, geostatistical modelling, spatial heterogeneity, or geographic co-risk.
C3	Biological interaction	Is there evidence that malaria and red blood cell traits interact biologically?	Evidence on HbAS protection, HbS selection, HbC, alpha-thalassaemia, G6PD, ABO, Dantu, epistasis, invasion, rosetting, or malaria severity modification.
C4	Clinical convergence	Do malaria and SCD converge clinically through shared syndromes or diagnostic ambiguity?	Evidence on severe anaemia, malaria-positive illness, transfusion need, febrile illness, respiratory distress, severe malaria phenotypes, or attribution.
C5	Service-mediated visibility	Does the health system shape whether	Evidence on screening, confirmatory

		SCD burden is recorded or hidden?	diagnosis, clinical recognition, routine reporting, coding, registers, mortality attribution, or hidden burden.
C6	Structural vulnerability	Does unequal access to essential services shape morbidity, mortality, recognition, or survival?	Evidence on access to screening, prophylaxis, vaccination, hydroxyurea, transfusion, specialist care, referral, retention, or follow-up.
C7	Programme response	Does the source describe or evaluate actions to identify, link, treat, refer, or retain people with SCD?	Evidence on newborn screening, point-of-care testing, early standardised care, screening-to-care linkage, referral systems, decentralisation, or implementation networks.

4. Code C1: Co-occurrence

Definition

Co-occurrence refers to evidence that malaria and SCD, HbS, HbAS, SCA, or malaria-relevant red blood cell polymorphisms occur in the same population, country, region, ecological setting, or clinical context.

Include when the source reports

1. Malaria and SCD in the same SSA population.
2. Malaria and HbS or HbAS in the same country or region.
3. Malaria and red blood cell polymorphisms in the same population.
4. SCD burden in malaria-endemic areas.
5. Malaria and SCD as overlapping public health problems.

Exclude when

1. The source discusses malaria alone with no SCD, HbS, red cell polymorphism, or severe anaemia relevance.
2. The source discusses SCD alone with no malaria-endemic context, malaria exposure, or SSA relevance.
3. The source mentions Africa only as a location without substantive relevance.
4. Co-occurrence is assumed by the reviewer but not reported or inferable from the study.

Example indicators

1. "Both malaria and SCD are common in the study region."

2. “The study was conducted in a malaria-endemic area with high HbS prevalence.”

3. “Children with SCD were studied in a malaria-endemic hospital setting.”

Reviewer note

Co-occurrence is the weakest syndemic-relevant category. It should not be interpreted as evidence of syndemic interaction unless the source also supports biological interaction, clinical convergence, service-mediated visibility, structural vulnerability, or programme response.

5. Code C2: Spatial co-risk

Definition

Spatial co-risk refers to evidence that malaria exposure, HbS frequency, SCD prevalence, SCA birth burden, child population need, or service access are geographically patterned, overlapping, or co-clustered.

Include when the source reports

1. Malaria prevalence, incidence, mortality, endemicity, or transmission maps.

2. HbS frequency maps.

3. SCD prevalence maps.

4. SCA birth-burden estimates with spatial distribution.

5. Subnational variation in sickle cell trait or SCD prevalence.

6. Spatial overlap between malaria exposure and HbS or SCD burden.

7. Facility catchments, geographic accessibility, travel time, or service distribution relevant to SCD care.

8. Spatial prioritisation for screening or service strengthening.

Exclude when

1. Geography is used only as a descriptive location.
2. A study reports national prevalence without spatial comparison or planning relevance.
3. A map is included but does not relate to malaria, HbS, SCD, severe anaemia, or service access.
4. Spatial evidence is unrelated to SSA or malaria-endemic African systems.

Example indicators

1. HbS frequency varies by district or region.
2. SCA births are estimated at subnational scale.
3. Malaria exposure is mapped across districts.
4. Facility access is modelled using travel time or catchment analysis.
5. Malaria burden and inherited sickle-related risk are mapped in the same country or region.

Sub-codes

Sub-code	**Label**	**Definition**
C2a	Malaria geography	Spatial distribution of malaria exposure, prevalence, incidence, or mortality.

C2b	HbS or SCD geography	Spatial distribution of HbS, SCD prevalence, or SCA births.
C2c	Overlap or co-risk	Evidence that malaria and HbS, SCD, or SCA burden overlap geographically.
C2d	Service geography	Spatial distribution of screening, diagnosis, transfusion, hydroxyurea, specialist care, referral, or follow-up.
C2e	Catchment relevance	Evidence at facility, district, or service-catchment scale.

Reviewer note

Spatial co-risk identifies potential convergence. It does not, by itself, prove clinical interaction, individual-level disease interaction, or service need.

6. Code C3: Biological interaction

Definition

Biological interaction refers to evidence that malaria and human red blood cell traits interact biologically, including through malaria protection, selection, parasite invasion, parasite growth, rosetting, cytoadherence, red cell membrane properties, immune mechanisms, or epistasis.

Include when the source reports

1. HbAS protection against severe malaria.
2. HbS persistence or malaria selection.

3. HbSS or SCA burden linked to HbS inheritance in malaria-endemic populations.
4. Malaria-relevant effects of HbC, alpha-thalassaemia, G6PD deficiency, ABO blood group, Dantu, or other red cell variants.
5. Parasite invasion or growth differences by genotype.
6. Rosetting, cytoadherence, or erythrocyte membrane mechanisms.
7. Co-inheritance or epistasis among malaria-relevant polymorphisms.
8. Genotype-specific malaria phenotypes or severity patterns.

Exclude when

1. The source discusses SCD biology without malaria relevance.
2. The source discusses malaria biology without red cell trait relevance.
3. The source is purely laboratory-based and does not inform human malaria-SCD interpretation.
4. The source mentions HbS or malaria without evidence of biological relationship.

Example indicators

1. HbAS reduces risk of severe *P. falciparum* malaria.
2. HbC is associated with reduced clinical malaria.
3. Blood group O reduces rosetting and severe malaria risk.

4. Dantu reduces parasite invasion through increased membrane tension.

5. HbAS and alpha-thalassaemia interact epistatically.

Sub-codes

Sub-code	**Label**	**Definition**
C3a	HbAS protection	Evidence that sickle cell trait modifies malaria risk or severity.
C3b	HbS selection	Evidence linking malaria ecology to HbS persistence.
C3c	Other polymorphisms	HbC, alpha-thalassaemia, G6PD, ABO, Dantu, or related variants.
C3d	Mechanistic pathway	Parasite invasion, growth, cytoadherence, rosetting, clearance, or immune mechanisms.
C3e	Epistasis	Co-inheritance or interaction between red cell polymorphisms.
C3f	Phenotype specificity	Different effects by malaria phenotype, such as severe malaria, severe malarial anaemia, cerebral malaria, or parasitaemia.

Reviewer note

Biological interaction is one of the strongest categories of syndemic-relevant evidence. It explains why malaria and SCD are historically and biologically linked. However, it does not explain service access, routine visibility, or programme response.

7. Code C4: Clinical convergence

Definition

Clinical convergence refers to evidence that malaria and SCD meet in clinical presentation, disease attribution, acute illness, or hospital care. This includes severe anaemia, malaria-positive illness, acute febrile illness, respiratory distress, transfusion need, severe malaria phenotypes, and diagnostic ambiguity.

Include when the source reports

1. Severe anaemia among children with SCD, SCA, HbSS, HbAS, malaria, or malaria-positive illness.

2. Malaria-positive illness among children with SCD or SCA.

3. Severe malaria phenotypes by genotype or SCD status.

4. Transfusion need in malaria-endemic settings where SCD is relevant.

5. Acute febrile illness or respiratory distress where malaria and SCD may overlap.

6. SCD under-recognition in severe anaemia, malaria, infection, or hospital admission categories.

7. Diagnostic attribution problems between malaria, SCD, severe anaemia, infection, and acute deterioration.

Exclude when

1. The source reports malaria clinical features without SCD, HbS, red cell polymorphism, or severe anaemia relevance.

2. The source reports SCD complications with no relevance to malaria-endemic settings, severe anaemia, or clinical convergence.

3. The clinical condition has no connection to malaria, SCD, severe anaemia, transfusion, or diagnostic attribution.

Example indicators

1. Children with SCA admitted with severe anaemia.

2. Malaria-positive children with SCA have different parasite density than HbAA children.

3. Severe anaemia admissions include previously unknown SCA.

4. Transfusion need occurs in children with severe anaemia where malaria and SCD overlap.

5. Malaria positivity complicates attribution of severe anaemia in children with SCA.

Sub-codes

Sub-code	Label	Definition
C4a	Severe anaemia	Severe anaemia as shared clinical syndrome.
C4b	Malaria-positive SCD illness	Malaria infection or positivity among children with SCD or SCA.

C4c	Severe malaria phenotype	Cerebral malaria, severe malarial anaemia, respiratory distress, or mixed severe malaria syndromes by genotype.
C4d	Transfusion-relevant illness	Evidence involving transfusion need or transfusion outcomes.
C4e	Diagnostic attribution	Uncertainty or misclassification between malaria, SCD, severe anaemia, infection, or other syndromes.
C4f	Under-recognised SCA in hospital care	Previously unknown SCA identified through testing or genotyping.

Reviewer note

Clinical convergence should be coded when malaria and SCD overlap in clinical presentation or interpretation. Do not require proof that malaria caused the illness. Diagnostic ambiguity itself is part of the clinical convergence domain.

8. Code C5: Service-Mediated Visibility

Definition

Service-mediated visibility refers to evidence that the health system shapes whether SCD burden becomes recorded, counted, linked to care, or hidden. It includes screening, confirmatory diagnosis, clinical recognition, hospital admission, diagnostic coding, routine reporting, clinic enrolment, referral, follow-up, and mortality attribution.

Include when the source reports

1. Newborn screening or infant testing that identifies SCD burden.
2. Confirmatory diagnosis or lack of confirmatory diagnosis.
3. Missed diagnosis, underdiagnosis, or under-recognition of SCD.
4. SCD identified through genotyping after routine clinical care.
5. Hospital records, clinic registers, laboratory registers, routine health information systems, HMIS, DHIS2, or surveillance data.
6. Mortality attribution or cause-of-death coding involving SCD.
7. Distinction between recorded burden and underlying burden.
8. Hidden SCD burden within severe anaemia, malaria, infection, respiratory disease, or mortality categories.

Exclude when

1. The source uses routine data but does not address diagnosis, reporting, coding, recognition, or visibility.
2. The source reports burden estimates without any relevance to service systems or visibility.
3. The source discusses health systems in general without SCD or malaria-SCD relevance.

Example indicators

1. Previously unknown SCA identified through post-trial genotyping.

2. SCD-attributable mortality exceeds cause-specific SCD mortality.
3. Newborn screening reveals SCD prevalence not visible in routine clinical data.
4. Routine data may reflect diagnostic access rather than true burden.
5. SCD is recorded under severe anaemia, malaria, infection, or respiratory illness.

Sub-codes

Sub-code	**Label**	**Definition**
C5a	Screening visibility	SCD becomes visible through newborn or infant screening.
C5b	Diagnostic visibility	Confirmatory testing or clinical recognition makes SCD visible.
C5c	Routine reporting	SCD visibility through HMIS, DHIS2, hospital records, clinic registers, or laboratory systems.
C5d	Mortality attribution	SCD captured or missed in cause-of-death systems.
C5e	Hidden burden	Evidence that SCD is undercounted, missed, or misclassified.
C5f	Service-visible burden	Recorded burden shaped by service contact and diagnostic capacity.

Reviewer note

Service-mediated visibility is not the same as structural vulnerability. Visibility concerns whether burden is recorded or hidden. Structural vulnerability concerns whether access conditions worsen outcomes, survival, or recognition. A study may receive both codes.

9. Code C6: Structural vulnerability

Definition

Structural vulnerability refers to evidence that unequal access to essential SCD services, resources, geography, health-system capacity, or social conditions modifies morbidity, mortality, recognition, or survival.

Include when the source reports

1. Limited access to newborn screening or early diagnosis.
2. Limited access to confirmatory testing.
3. Limited access to caregiver counselling.
4. Limited access to penicillin prophylaxis or vaccination.
5. Limited access to hydroxyurea.
6. Limited access to safe blood transfusion or transfusion readiness.
7. Limited access to specialist SCD care.
8. Referral barriers, transport barriers, cost barriers, workforce gaps, laboratory gaps, or medicine supply gaps.

9. Retention or follow-up challenges.

10. Mortality, morbidity, or invisibility linked to service access limitations.

Exclude when

1. The source discusses poverty or social determinants without SCD, malaria, severe anaemia, or care access relevance.

2. The source reports clinical outcomes without examining or implying structural access conditions.

3. The source describes a programme but does not address access, barriers, or service capacity.

Example indicators

1. Children with SCD lack access to newborn screening.

2. Hydroxyurea is effective but unavailable or difficult to sustain.

3. Transfusion capacity limits management of severe anaemia.

4. Specialist SCD care is concentrated in tertiary facilities.

5. Children are lost after screening because of weak referral or follow-up systems.

Sub-codes

Sub-code	Label	Definition
C6a	Diagnostic access	Access to screening or confirmatory diagnosis.

C6b	Preventive care access	Penicillin, vaccination, malaria prevention, caregiver education.
C6c	Disease-modifying therapy access	Hydroxyurea or other disease-modifying care.
C6d	Acute care access	Transfusion readiness, emergency care, severe anaemia care.
C6e	Specialist and referral access	Specialist clinics, referral networks, hub-and-spoke systems.
C6f	Retention and continuity	Follow-up, clinic retention, long-term care.
C6g	Structural barriers	Cost, distance, transport, workforce, laboratory systems, blood supply, medicines.

Reviewer note

Structural vulnerability should be coded when access conditions shape risk, outcome, recognition, or survival. It is stronger when the study reports barriers, service gaps, mortality, or care discontinuity.

10. Code C7: Programme response

Definition

Programme response refers to evidence on actions, models, interventions, or systems designed to identify, diagnose, enrol, treat, refer, or retain people with SCD in SSA.

Include when the source reports

1. Newborn screening programmes.
2. Infant screening or surveillance platforms.
3. Point-of-care testing.
4. Confirmatory diagnosis pathways.
5. Caregiver notification and counselling.
6. Clinic enrolment and follow-up.
7. Early standardised care.
8. Penicillin prophylaxis, vaccination, hydroxyurea, or transfusion as programme components.
9. Decentralised care or hub-and-spoke models.
10. Referral systems.
11. Implementation networks, such as multicountry screening initiatives.
12. Programme sustainability, financing, scale-up, or service strengthening.

Exclude when

1. The source describes SCD treatment without programme, implementation, or service-system relevance.

2. The source describes a health programme unrelated to SCD, malaria, screening, or care linkage.

3. The source mentions screening without reporting implementation, linkage, or programme relevance.

Example indicators

1. Ghana newborn screening programme.

2. Uganda dried blood spot surveillance for SCD.

3. CONSA newborn screening implementation model.

4. Point-of-care test evaluation for SCD.

5. Screening-to-care cascade with confirmation, notification, enrolment, prophylaxis, and follow-up.

Sub-codes

Sub-code	**Label**	**Definition**
C7a	Screening programme	Newborn, neonatal, infant, or population screening.
C7b	Diagnostic innovation	Point-of-care testing or laboratory systems.
C7c	Linkage to care	Result return, confirmation, notification, counselling, enrolment.
C7d	Early standardised care	Prophylaxis, vaccination, malaria prevention,

		hydroxyurea assessment, monitoring.
C7e	Referral and networked care	Decentralised care, hub-and-spoke systems, referral pathways.
C7f	Implementation network	Multicountry or national implementation collaborations.
C7g	Sustainability and scale-up	Financing, policy, workforce, systems integration, long-term viability.

Reviewer note

Programme response is action-oriented. It should be coded when the source reports a programme, intervention, implementation model, diagnostic platform, care pathway, or service system.

11. Explicit Syndemic theory code

Definition

Explicit syndemic theory refers to direct use of syndemic terminology, syndemic theory, or formal syndemic concepts.

Include when the source

1. Uses the word “syndemic” or “syndemics.”

2. Cites syndemic theory.

3. Frames malaria and SCD, or related conditions, as a syndemic.

4. Tests or discusses syndemic interaction, clustering, structural amplification, or excess burden using syndemic theory.

Exclude when

1. The source describes co-occurrence but does not use syndemic language or concepts.

2. The source uses terms like comorbidity, overlap, or interaction without syndemic framing.

3. The source uses syndemic language in an unrelated disease context.

Reviewer note

Explicit syndemic theory is expected to be uncommon. Do not code a source as explicit syndemic theory unless syndemic terminology or theory is clearly present.

12. Implicit syndemic logic code

Definition

Implicit syndemic logic refers to evidence that does not use syndemic terminology but supports syndemic reasoning through links between disease clustering, biological interaction, clinical convergence, structural vulnerability, service-mediated visibility, or programme response.

Include when the source

1. Links malaria and SCD through biological, clinical, spatial, or health-system pathways.

2. Shows that structural conditions shape outcomes or visibility.

3. Demonstrates under-recognition of SCD within clinical or routine categories.

4. Links screening, diagnosis, treatment, referral, and follow-up as a programme pathway.

5. Provides evidence for more than one syndemic-relevant category.

Exclude when

1. The source reports a single condition with no connection to another domain.

2. The source provides descriptive co-occurrence only.

3. The reviewer must speculate beyond what the source reports.

Reviewer note

Implicit syndemic logic should be coded carefully. It is strongest when the study connects at least two domains.

13. Level of evidence integration

Each included source should be assigned one integration level.

Table 2. Evidence integration levels

Level	**Label**	**Definition**	**Example**
Level 1	Single-domain evidence	The source contributes evidence to one domain only.	A study of HbAS protection against malaria.
Level 2	Two-domain evidence	The source links two domains.	A study linking HbS geography to malaria endemicity.
Level 3	Multi-domain evidence	The source links three or more domains.	A study linking SCA, severe anaemia, malaria positivity, and under-recognition.
Level 4	Syndemic-integrated evidence	The source links biological interaction, spatial co-risk,	An integrated planning study linking malaria-SCD

		clinical convergence, service-mediated visibility, structural vulnerability, and programme response.	co-risk, service readiness, routine visibility, and screening-to-care cascade outcomes.

Coding guidance

1. Use Level 1 when the study contributes to one evidence category only.
2. Use Level 2 when the study clearly links two categories.
3. Use Level 3 when three or more categories are connected.
4. Use Level 4 only when the study provides an integrated syndemic or programme-planning framework connecting multiple levels.

14. Strength of syndemic relevance

In addition to domain codes, reviewers should judge the strength of syndemic relevance.

Table 3. Syndemic relevance rating

Rating	Definition	Coding rule
Low	Evidence shows co-occurrence or background relevance only.	Usually C1 only.
Moderate	Evidence links two domains or shows spatial, biological, clinical, visibility, structural, or programme relevance.	Usually Level 2.
High	Evidence links three or more domains and supports a	Usually Level 3.

	pathway between malaria and SCD.	
Very high	Evidence provides integrated analysis across biological, spatial, clinical, visibility, structural, and programme domains.	Usually Level 4.

Reviewer note

Syndemic relevance is not the same as study quality. A well-conducted single-domain study may have low syndemic integration but high scientific importance.

15. Handling ambiguous sources

15.1 If a source mentions malaria and SCD but does not link them

Code C1 only if the shared context is meaningful. Do not code interaction.

15.2 If a source maps malaria but not HbS or SCD

Code C2 only if the malaria geography is relevant to co-risk interpretation or later integration.

15.3 If a source studies SCD care in SSA but does not mention malaria

Code C6 or C7 if it addresses service access or programme response in malaria-endemic SSA. Do not code biological interaction or clinical convergence unless reported.

15.4 If a source reports severe anaemia but not SCD testing

Code C4 only if severe anaemia is relevant to malaria-SCD convergence, under-recognition, or attribution. If relevance is unclear, note uncertainty.

15.5 If a source is a review

Code based on the evidence domains covered in the review. Do not automatically assign all codes unless the review substantively discusses each domain.

16. Coding Examples

Table 4. Example Coding Scenarios

Source type	Likely codes	Integration level	Notes
HbAS protection cohort study	C3	Level 1	Strong biological evidence, limited service relevance.
HbS geostatistical mapping study	C2, C3	Level 2	Links spatial risk and biological selection.
Malaria map without SCD or HbS	C2	Level 1	Include only if relevant to malaria-SCD co-risk.
Severe anaemia trial with SCA genotyping	C4, C5	Level 2 or 3	Clinical convergence and visibility.
Newborn screening programme	C5, C6, C7	Level 2 or 3	Visibility, structural access, programme response.
Hydroxyurea trial in SSA	C6, C7	Level 2	Essential care and programme relevance.
GBD SCD mortality attribution paper	C5, C6	Level 2	Mortality visibility and burden.
Integrated spatial planning study	C2, C5, C6, C7	Level 3 or 4	Strong planning relevance.
Syndemic theory paper	Explicit syndemic theory	Level depends on application	Use to support conceptual framework.

17. Reviewer justification field

For each included source, reviewers should complete a brief justification statement.

Suggested format

"This source was coded as [codes] because it reports [key evidence]. It was assigned integration level [level] because it connects [domains]."

Example

"This source was coded as clinical convergence and service-mediated visibility because it reports previously unknown SCA among children admitted with severe anaemia and compares malaria biomarkers by haemoglobin genotype. It was assigned Level 3 because it connects genotype, clinical presentation, malaria positivity, and diagnostic recognition."

18. Quality control

To maintain coding consistency:

1. Reviewers will pilot the codebook on a sample of included studies.
2. Disagreements will be discussed by the review team.
3. Coding definitions will be revised if repeated ambiguity occurs.
4. All changes will be documented in a version-control log.
5. Final codes will be reviewed during synthesis for consistency across domains.

19. Version control log

Version	**Date**	**Change made**	**Reason**	**Approved by**
1.0	To be inserted	Initial codebook	Initial version	To be inserted
1.1	To be inserted	To be inserted	To be inserted	To be inserted